M13 Batch 2 — Full Set

M13.3 Autoflight (ATA 22) · PDF pages 64–155

Control engineering and flight guidance fundamentals, autopilot engagement and interlocks, control surfaces, pitch modes and trim, roll and heading modes, cross coupling.

306 questions · Level 3: 109 · Level 2: 143 · Level 1: 54

● marks a question built on a fact your instructor highlighted or underlined in the manual (276 of 306). Topics run in manual page order; inside each topic the questions are grouped by level, and the answers with explanations follow at the end of that topic.

No calculations, no unit conversions, no worked examples, nothing type-specific unless it was marked.

Introduction and control engineering fundamentals

PDF pages 68–68 · 10 questions

LEVEL 1 — RECALL

1. Which other name is given to closed loop control ●
   a. actuation
   b. correction
   c. regulation

2. Which quantity is denoted by Xw in the terms used for closed loop control ●
   a. control value
   b. reference value
   c. error

LEVEL 2 — UNDERSTAND AND EXPLAIN

3. Besides permanent monitoring of the control value, which feature is typical of closed loop control ●
   a. its chain like sequence of effects through the actuator
   b. its feedback to and comparison with the reference value
   c. its conversion by the controller into an input value

4. Which feature is characteristic of open loop control ●
   a. the comparison made at the controller input
   b. the chain like sequence of effects
   c. the permanent monitoring of the output

5. In a closed loop system the control value and the reference value are found to differ. What happens to that difference? ●
   a. It is passed to the measurement unit and stored until the next comparison
   b. It is converted in the actuator into an output which acts on the controller
   c. It is converted in the controller into an output which acts on the actuator

6. In a closed loop system, which quantity of the controlled system does the actuator modify ●
   a. the disturbance acting on it
   b. its output, the control value
   c. its input, the reference value

7. In an open loop arrangement, on what does the controller output act? ●
   a. On the system to be controlled, which in turn acts on an actuator
   b. On a measurement unit, which in turn acts on the reference value
   c. On an actuator, which in turn acts on the system to be controlled

LEVEL 3 — APPLY AND REASON

8. An open loop system holds its output value constant while a disturbance acts; the reference value is varied until which condition is reached ●
   a. the disturbance is measured and fed back to the controller input for comparison
   b. the controller output and the reference value become equal to each other
   c. the effect of the controller output and that of the disturbance equalize each other

9. A controlled system is acted on by disturbances whose effect cannot be predicted. Why is open loop control the wrong choice here? ●
   a. It needs a measurement unit that unpredictable disturbances overload
   b. It feeds the disturbance back and so amplifies the error further
   c. It can only correct for what was allowed for in advance

10. A gust disturbs an aircraft that is being regulated. How is that disturbance dealt with? ●
   a. It is fed back to the controller input and so more or less equalized
   b. It is removed by the actuator before it reaches the controlled system
   c. It is predicted by the controller and allowed for before it arrives

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Answers — Introduction and control engineering fundamentals

1. **c** Closed loop control is also called regulation, because the control value is permanently monitored and compared with the reference value. (L1, PDF p. 68)
2. **c** Xw is the symbol given for the error in a closed loop system. (L1, PDF p. 68)
3. **b** Closed loop control is defined by feeding the control value back and comparing it with the reference value. (L2, PDF p. 68)
4. **b** In open loop control the effects run in a chain from input to output with no return path. (L2, PDF p. 68)
5. **c** The error goes to the controller, whose output drives the actuator. (L2, PDF p. 68)
6. **b** The actuator acts on the controlled system so that the control value changes. (L2, PDF p. 68)
7. **c** The order runs controller, then actuator, then controlled system. (L2, PDF p. 68)
8. **c** With no feedback the only way to hold the output is to trim the reference until the two effects cancel. (L3, PDF p. 68)
9. **c** Without feedback the correction must be worked out beforehand, which an unpredictable disturbance makes impossible. (L3, PDF p. 68)
10. **a** Feedback carries the disturbance back to the controller input, where it is largely cancelled. (L3, PDF p. 68)

Regulator types, purpose and tasks of flight guidance

PDF pages 72–78 · 44 questions

LEVEL 1 — RECALL

11. For a P - Regulator, to what is the change in the output value always proportional ●

- a. the rate of change of the input value
- b. the change in the input value
- c. the magnitude of the input value

12. For an I - Regulator, to what is the rate of change of the output value proportional ●

- a. the change in its output value
- b. its input value
- c. the rate of change of its input value

13. How are the six degrees of freedom of an aircraft in the air divided ●

- a. four rotary and two translatory degrees of freedom
- b. three rotary and three translatory degrees of freedom
- c. two rotary and four translatory degrees of freedom

14. Which axes do the three degrees of rotary freedom turn about ●

- a. roll, pitch and yaw axes
- b. longitudinal, lateral and flight path axes
- c. roll, pitch and heading axes

15. What do flight parameters describe ●

- a. the state of the aircraft fixed coordinate system
- b. the state of motion of an aircraft
- c. the state of the autopilot regulators

16. Which motions are described by flight condition parameters ●

- a. translatory motions
- b. rotary motions
- c. oscillatory motions

17. Into which two parts is the flight path of a transport category airplane separated ●

- a. an inner and an outer profile
- b. a horizontal and a vertical profile
- c. a rotary and a translatory profile

18. Which autopilot channel is used to follow a vertical profile ●

- a. longitudinal or pitch channel
- b. directional or yaw channel
- c. lateral or roll channel

LEVEL 2 — UNDERSTAND AND EXPLAIN

19. A control loop contains one energy storage device. What does this produce? ●

- a. An output value that follows the input value without any delay
- b. An oscillation of the output value after a change of the input
- c. A delay in the development of the output value over time

20. A control loop contains two energy storage devices. What is the consequence?

- **a.** Energy is held fast, so the output value is merely delayed
- **b.** Energy is dissipated, so the output value follows without delay
- **c.** Energy can flow to and fro, so the output value oscillates

21. Under which condition does a P - Regulator generate a controller output value

- **a.** when an error value exists at its input
- **b.** when the error value at its input is zero
- **c.** when the input value is changing at a constant rate

22. A constant positive input value is applied to an I - Regulator. What does its output do?

- **a.** It decreases linearly
- **b.** It stays at its present magnitude
- **c.** It increases linearly

23. The input value of an I - Regulator falls to zero. What becomes of its output value?

- **a.** It decreases linearly from its present magnitude
- **b.** It falls to zero together with the input value
- **c.** It remains constant at its present magnitude

24. The input of a D - Regulator changes at a constant rate. What does the output value do?

- **a.** It remains steady at one level
- **b.** It falls back towards zero
- **c.** It increases linearly without limit

25. What are the three degrees of translatory freedom

- **a.** the linear movements along the X, Y and Z axes
- **b.** the linear movements along the roll and pitch axes
- **c.** the rotations about the X, Y and Z axes

26. Why does the pilot have two areas of responsibility at the same time?

- **a.** Because the aircraft has two different coordinate systems defined
- **b.** Because the aircraft has two different kinds of degree of freedom
- **c.** Because the aircraft has two separate sets of flight controls

27. Which degrees of freedom are affected by the control inputs that stabilize the aircraft attitude about its centre of gravity

- **a.** translatory degrees of freedom
- **b.** navigation objectives
- **c.** rotary degrees of freedom

28. Why is relieving the crew of monotonous flight path guidance tasks worthwhile?

- **a.** Monotony deteriorates and limits the attention they pay to monitoring systems
- **b.** Monotony makes the autopilot less accurate in holding the selected values
- **c.** Monotony prevents the crew from entering reference values on the panel

29. What can bring a human being to the limit of mental capacity in difficult flight phases

- **a.** a multitude of tasks which have to be carried out concurrently
- **b.** a multitude of disturbances which have to be predicted in advance
- **c.** a multitude of reference values which have to be entered in sequence

30. Which direction is positive for the lateral axis in the aircraft fixed coordinate system ●

- **a.** in the port direction
- **b.** in the starboard direction
- **c.** in the direction of flight

31. Which direction is positive for the vertical axis in the aircraft fixed coordinate system ●

- **a.** downwards
- **b.** upwards
- **c.** in the direction of flight

32. For auto flight purposes the body fixed coordinate axes are given a second set of names. What do those names refer to? ●

- **a.** Because they refer to the linear motion along the respective axis
- **b.** Because they refer to the rotary motion around the respective axis
- **c.** Because they refer to the position of the centre of gravity

33. How is a predetermined condition or path realized in flight condition and flight path regulation ●

- **a.** by permanently adapting the reference values to one or several flight parameters
- **b.** by permanently selecting a new mode of operation for each flight parameter
- **c.** by permanently adapting one or several flight parameters to their reference values

34. For which systems is the conduct of flight decomposed into part-tasks in automatic flight guidance ●

- **a.** the flight management system and the auto-throttle / auto-thrust system
- **b.** the autopilot / flight director and the flight management system
- **c.** the autopilot / flight director and the auto-throttle / auto-thrust system

35. What decides which flight parameters are controlled in a given flight phase? ●

- **a.** The reference values entered for the AP/FD and the ATHR
- **b.** The mode of operation selected for the AP/FD and the ATHR
- **c.** The degree of automation achieved in that autoflight system

36. On which other mode of operation does that of the auto-throttle / auto-thrust system depend ●

- **a.** the autopilot's longitudinal channel
- **b.** the autopilot's lateral channels
- **c.** the flight director's roll channel

37. Besides controlling air speed, which other task is given to the ATHR system ●

- **a.** setting a flight phase dependant altitude target
- **b.** setting a flight phase dependant pitch attitude
- **c.** setting a flight phase dependant thrust rating

38. During climb, cruise and descent, how are the modes of the longitudinal channel and of the lateral channel chosen? ●

- **a.** They are selected in a strict sequence by the system
- **b.** They are selectable independently from each other
- **c.** They are selected together as a fixed combination

39. In a modern autoflight system, what can be said about the operating modes offered to the flight director? ●

- **a.** A separate and shorter list of modes is offered for it
- **b.** Only the basic modes of the autopilot can be selected for it
- **c.** Most of the modes available for the autopilot can be selected for it too

40. How is the provision of reference values and the tuning of the VOR / ILS receivers allocated among the part-tasks ●

- **a.** It belongs to the auto-thrust part-task
- **b.** It belongs to each of the part-tasks
- **c.** It belongs to the flight director part-task

LEVEL 3 — APPLY AND REASON

41. The output of a control loop is found to oscillate after every input change. What must be provided and why? ●

- **a.** A second energy store, because the loop then settles faster
- **b.** Damping, because without it the loop will not be stable
- **c.** A larger error value, because the controller then responds

42. Why can a P - Regulator never completely compensate a change of the reference value or a disturbance? ●

- **a.** Its output is proportional to how fast the error value is changing
- **b.** Inaccuracies may show an error of zero while a real difference remains
- **c.** Its output becomes infinite whenever the error value reaches zero

43. Which regulator can produce a controller output from a very small error that persists for a sufficient period of time ●

- **a.** P - Regulator, which can completely compensate the change
- **b.** D - Regulator, which can completely compensate the change
- **c.** I - Regulator, which can completely compensate the change

44. The input of a D - Regulator leaps from one constant value to another. What is the theoretical result at the output? ●

- **a.** It becomes constant
- **b.** It becomes zero
- **c.** It becomes infinite

45. Which regulator behaviour is used when an error just beginning to build up has to be removed very quickly ●

- **a.** D - Regulator, whose output answers the rate of change
- **b.** P - Regulator, whose output answers the rate of change
- **c.** I - Regulator, whose output answers the rate of change

46. With one exception, direct inputs affecting the translatory degrees of freedom are not possible. How then is the flight path controlled? ●

- **a.** Indirectly, through inputs affecting initially the rotary degrees of freedom
- **b.** Directly, through inputs applied to each of the three translatory axes
- **c.** Indirectly, through inputs affecting initially the centre of gravity position

47. Along which axis does engine thrust, the one direct input affecting a translatory degree of freedom, act ●

- **a.** lateral or Y - axis
- **b.** vertical or Z - axis
- **c.** longitudinal or X - axis

48. An approach is to be flown in very low visibility. Why must the autoflight system fly it? ●

- **a.** The thrust of the engines cannot be set manually in such conditions
- **b.** Regulations or limited perception prevent manual control in such phases
- **c.** The flight director cannot supply commands in such conditions

49. Why can the flight condition task and the flight path task not be clearly distinguished from each other in every case? ●

- **a.** Any control of flight condition parameters uses the same regulator type
- **b.** Any control of flight condition parameters always affects the flight path too
- **c.** Any control of flight path parameters is carried out by the inner loop too

50. Which flight parameter influences the flight condition as well as the flight path and so blurs the division between the two tasks ●

- **a.** the roll attitude
- **b.** the airspeed
- **c.** the pitch attitude

51. What makes it possible to guide a flight automatically from the initial climb out until the roll out after landing? ●

- **a.** Sequencing the different control tasks with respect to the required flight phases
- **b.** Selecting a single control task that covers every one of the flight phases
- **c.** Adapting the reference values to the control values in every flight phase

52. A quantity describing a linear motion of the aircraft is to be regulated. Which part of the autopilot has this task? ●

- **a.** The inner loop, since the quantity is a flight path parameter
- **b.** The outer loops, since the quantity is a flight path parameter
- **c.** The outer loops, since the quantity is a flight condition parameter

53. A precision approach and landing is made up of a strict sequence of operating modes needing a series of different control parameters. What follows from this? ●

- **a.** Mode sequencing is inhibited, so the approach mode holds to roll out
- **b.** Mode sequencing is automatic, so one button relieves the pilot to roll out
- **c.** Mode sequencing is manual, so the pilot must select each mode in turn

54. Why is air speed monitored by the activated autoflight system to prevent upper and lower limits being exceeded? ●

- **a.** Air speed is a flight condition parameter of the rotary freedoms
- **b.** Air speed is the only parameter the auto-thrust system can control
- **c.** Air speed is a decisive criterion for the amount of lift produced

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Answers — Regulator types, purpose and tasks of flight guidance

11. **b** A proportional regulator ties the change of its output directly to the change of its input. (L1, PDF p. 72)

12. **b** In an integral regulator it is how fast the output moves that follows the input. (L1, PDF p. 72)

13. **b** The six split evenly into three rotary and three translatory. (L1, PDF p. 74)

14. **a** Rotation takes place about the X or roll, Y or pitch and Z or yaw axes. (L1, PDF p. 74)

15. **b** A flight parameter is a physical quantity that describes how the aircraft is moving. (L1, PDF p. 76)

16. **b** Flight condition parameters belong to the rotary degrees of freedom. (L1, PDF p. 76)

17. **b** The path is split into a horizontal profile and a vertical profile. (L1, PDF p. 78)

18. **a** The vertical profile is flown by the longitudinal, that is the pitch, channel. (L1, PDF p. 78)

19. **c** One storage device gives P1 behaviour, which is a lag in the output. (L2, PDF p. 72)

20. **c** Two storage devices let energy pass back and forth, which shows up as oscillation. (L2, PDF p. 72)

21. **a** No error at the input means no output from a proportional regulator. (L2, PDF p. 72)

22. **c** With a steady positive input the integral regulator's output ramps up at a steady rate. (L2, PDF p. 72)

23. **c** A zero input means the output stops moving but holds the level it has reached. (L2, PDF p. 72)

24. **a** The differential regulator reacts to rate of change, so a steady rate gives a steady output. (L2, PDF p. 72)
25. **a** Translatory freedom is straight line movement along each of the three axes. (L2, PDF p. 74)
26. **b** The rotary and translatory freedoms give rise to two separate jobs for the pilot. (L2, PDF p. 74)
27. **c** Attitude about the centre of gravity is a rotary matter. (L2, PDF p. 74)
28. **a** Dull repetitive tasks eat into the attention the crew has left for watching the systems. (L2, PDF p. 74)
29. **a** It is the number of jobs that have to be done at the same time that over-stresses the pilot. (L2, PDF p. 74)
30. **b** The lateral axis is counted positive to the right, that is to starboard. (L2, PDF p. 74)
31. **a** The vertical axis is taken positive towards the ground. (L2, PDF p. 74)
32. **b** Each axis is named after the rotation that takes place about it. (L2, PDF p. 74)
33. **c** The control values are continuously driven towards the reference values that were set. (L2, PDF p. 76)
34. **c** The part-tasks are shared between the AP/FD and the ATHR. (L2, PDF p. 78)
35. **b** Selecting a mode of operation is what fixes the parameters to be controlled. (L2, PDF p. 78)
36. **a** The thrust system follows whatever the vertical channel of the autopilot is doing. (L2, PDF p. 78)
37. **c** Its other job is to set the thrust rating appropriate to the flight phase. (L2, PDF p. 78)
38. **b** In these phases the vertical and lateral modes are picked separately. (L2, PDF p. 78)
39. **c** Modern systems let nearly the whole autopilot mode list be used by the flight director. (L2, PDF p. 78)
40. **b** Every part-task includes setting the reference values and tuning the navaids. (L2, PDF p. 78)
41. **b** Oscillation around the reference value is P2 behaviour, and damping is mandatory if the closed loop is to be stable. (L3, PDF p. 72)
42. **b** A proportional regulator needs an error to act, and measurement inaccuracy can hide a genuine one. (L3, PDF p. 72)
43. **c** The integral regulator acts on tiny lasting errors and so removes the offset entirely. (L3, PDF p. 72)
44. **c** A step is an infinitely fast change, so a rate sensitive regulator gives an infinite output. (L3, PDF p. 72)
45. **a** The differential regulator sees the error growing and reacts at once, which is its stated advantage. (L3, PDF p. 72)
46. **a** The pilot has to change attitude first and let the path follow from it. (L3, PDF p. 74)
47. **c** Engine thrust pushes the aircraft straight along its longitudinal axis. (L3, PDF p. 74)
48. **b** In these phases the rules, or simply what the pilot can see, rule manual control out. (L3, PDF p. 74)
49. **b** Changing attitude inevitably moves the aircraft, so the two tasks overlap. (L3, PDF p. 76)
50. **b** Airspeed is named as the parameter that belongs to both tasks. (L3, PDF p. 76)
51. **a** Automatic guidance over a whole flight comes from running the control tasks in the order the phases need. (L3, PDF p. 76)
52. **b** Linear motion means a flight path parameter, and flight path regulation is the outer loop task. (L3, PDF p. 76)
53. **b** Because the modes must follow a fixed order, the system runs them itself after a single selection. (L3, PDF p. 78)
54. **c** Lift depends on speed, so the speed envelope has to be protected. (L3, PDF p. 78)

Flight plan structure and the autoflight system

PDF pages 80–84 · 25 questions

LEVEL 1 — RECALL

55. Which flight phases make up the vertical structure of a flight plan ●

- a. departure, climb, cruise, arrival and approach
- b. take-off, climb, cruise, descent and landing
- c. take-off, climb, cruise, descent and approach

56. What are the segments making up the lateral structure of a flight plan called ●

- a. tracks
- b. fixes
- c. legs

57. Into how many categories are all legs of a flight plan subdivided for procedural reasons ●

- a. three categories
- b. four categories
- c. five categories

58. Which subsystems does the autoflight system comprise ●

- a. autopilot, flight director and auto-throttle / auto-thrust system
- b. autopilot, flight director and flight management computer system
- c. autopilot, flight management system and auto-throttle / auto-thrust system

LEVEL 2 — UNDERSTAND AND EXPLAIN

59. What essentially determines the vertical flight path guidance written into a flight plan? ●

- a. The performance of the aircraft
- b. The intended route of the aircraft
- c. The air traffic control clearances

60. Besides air speed and thrust rating, what characterizes each vertical phase ●

- a. heading, track angle or wind correction angle
- b. pitch attitude, roll attitude or yaw attitude
- c. altitude, climb or sink rate or flight path angle

61. Which document describes both the vertical and the lateral flight path guidance for a flight from take-off to touch down ●

- a. flight plan
- b. flight parameter list
- c. flight profile

62. Besides its direction, what defines a leg of a flight plan ●

- a. the distance between two waypoints
- b. the heading between two waypoints
- c. the time between two waypoints

63. Where do the legs of the standard instrument departure end ●

- a. at the initial approach fix
- b. at the airway system
- c. at the active runway

64. Which legs does the standard terminal arrival route comprise ●

- **a.** from the initial approach fix to the active runway
- **b.** from the airway system to the initial approach fix
- **c.** from the initial approach fix to the airway system

65. Which other legs are taken in by the leg category that ends at the active runway ●

- **a.** all the legs of the departure route
- **b.** all the legs of a missed approach
- **c.** all the legs of the arrival route

66. Flying leg by leg, what must the pilot take into consideration when he selects the heading for each leg? ●

- **a.** The initial approach fix
- **b.** The flight path angle
- **c.** The wind correction angle

67. Which position sensors does the FMCS use to provide positioning and navigation automatically ●

- **a.** DME/DME, VOR/ILS, IRS and ATC
- **b.** VOR/DME, ILS, radio altimeter and GPS
- **c.** DME/DME, VOR/DME, IRS and GPS

68. Which two tasks does the pilot have to fulfil during a flight

- **a.** flight management and flight condition damping
- **b.** flight management and flight path control
- **c.** flight path control and flight mode annunciation

69. Besides separately, how may each regulator offered by the autoflight system be used ●

- **a.** in combination with the flight management computer system
- **b.** in combination with the measurement unit of the control loop
- **c.** in combination with the auto-throttle / auto-thrust system

LEVEL 3 — APPLY AND REASON

70. Which parts of the system provide the combinations of regulators used for automatic vertical flight path guidance? ●

- **a.** The AP/FD pitch channel and the flight management system under ATC control
- **b.** The AP/FD roll channel and the auto-thrust system under FMCS control
- **c.** The AP/FD pitch channel and the auto-thrust system under AP/FD control

71. The roll channel of the autopilot/flight director is working as an independent subsystem. Why can it not guide the aircraft along a horizontal profile? ●

- **a.** It needs inputs from the pilot or from a flight management computer system
- **b.** It needs the pitch channel to be engaged in a vertical profile mode
- **c.** It needs the auto-thrust system to set a flight phase thrust rating

72. A waypoint is co-located with a VOR navaid. What does this allow the pilot to do? ●

- **a.** Select the appropriate autopilot approach mode to capture the leg
- **b.** Leave the roll channel to track the leg without any further input
- **c.** Select the appropriate autopilot NAV mode to track a leg to or from it

73. Why is leg by leg guidance described as semi-automatic flight guidance only? ●

- **a.** New parameters must be set up by the pilot before each following leg
- **b.** The autopilot roll channel must be disengaged on each following leg
- **c.** New parameters must be set up by the FMCS before each following leg

74. What becomes possible once every input the AP/FD roll and yaw channel needs is provided for it ●

- **a.** shortly after take-off the aircraft can be guided along the flight plan automatically
- **b.** shortly after take-off the roll channel can be operated as an independent subsystem
- **c.** shortly before each waypoint the pilot can set up the parameters for the next leg

75. The accuracy of each sensor is permanently monitored. What does the FMCS then do with the information they give? ●

- **a.** It blends it to calculate an accurate position in the earth coordinate system
- **b.** It selects the single most accurate sensor and discards the other information
- **c.** It passes it unchanged to the roll channel as a wind correction angle

76. Which navigation parameters can the flight management computer system calculate from the change of the aircraft position ●

- **a.** Mach number or flight path angle
- **b.** air speed or heading
- **c.** ground speed or track

77. A flight is composed of a large number of very different route segments. What does the autoflight system therefore make available?

- **a.** An appropriate regulator or combination of regulators for each route segment
- **b.** An appropriate reference value or combination of values for each route segment
- **c.** An appropriate position sensor or combination of sensors for each route segment

78. What is needed before extensive automatic flight guidance, a managed automatic flight, becomes possible?

- **a.** Interaction of the flight management system with the autoflight system
- **b.** Interaction of the flight director with the autopilot roll channel
- **c.** Interaction of the autopilot with the auto-throttle system alone

79. Particular route segments feature very different flight parameters. How may the pilot deal with a segment he does not wish to fly by hand? ●

- **a.** Delegate the task to the flight director working with the auto-thrust system
- **b.** Delegate the task to the flight management system working with the flight director
- **c.** Delegate the task to the autopilot working with the auto-thrust system

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Answers — Flight plan structure and the autoflight system

55. **c** The five vertical phases listed end with approach. (L1, PDF p. 80)

56. **c** The segments that make up the lateral structure are called legs. (L1, PDF p. 82)

57. **b** There are four leg categories. (L1, PDF p. 82)

58. **a** The three named subsystems make up the autoflight system. (L1, PDF p. 84)

59. **a** What the aircraft can do sets the vertical profile. (L2, PDF p. 80)

60. **c** Besides speed and thrust rating, each phase is set by altitude, climb or sink rate, or flight path angle. (L2, PDF p. 80)

61. **a** It is the flight plan that sets out both the vertical and lateral guidance. (L2, PDF p. 80)

62. **a** Distance between two waypoints plus direction is what defines a leg. (L2, PDF p. 82)

63. **b** The SID runs from the start of the flight plan up to the airway system. (L2, PDF p. 82)

64. **b** The STAR takes the aircraft off the airways down to the initial approach fix. (L2, PDF p. 82)

65. **b** The approach category also takes in the missed approach legs. (L2, PDF p. 82)

66. **c** Heading for a leg must allow for the wind correction angle. (L2, PDF p. 82)

67. **c** Those four sensor sources are the ones used for FMCS positioning. (L2, PDF p. 82)

68. **b** The pilot's two tasks are flight management and flight path control. (L2, PDF p. 84)

69. **c** A regulator can work alone or together with the thrust system. (L2, PDF p. 84)

70. **c** Vertical guidance comes from the pitch channel together with the thrust system under AP/FD control. (L3, PDF p. 80)

71. **a** On its own the roll channel has nothing to steer to, so it must be fed by the pilot or the FMCS. (L3, PDF p. 82)

72. **c** A waypoint on a navaid lets the NAV mode of the autopilot track the leg. (L3, PDF p. 82)

73. **a** The pilot has to reset the parameters at every waypoint, so the guidance is only half automatic. (L3, PDF p. 82)

74. **a** With the FMCS feeding the lateral channels, automatic guidance can begin soon after take-off. (L3, PDF p. 82)

75. **a** The FMCS mixes all the position sources into one accurate FMS position. (L3, PDF p. 82)

76. **c** Watching how the position moves gives ground speed and track. (L3, PDF p. 82)

77. **a** Because the segments differ so much, a matching regulator or set of regulators is offered for each. (L3, PDF p. 84)

78. **a** Managed automatic flight comes only from the two systems working together. (L3, PDF p. 84)

79. **c** Handing the segment to the autopilot and auto-thrust system gives automatic guidance for that phase. (L3, PDF p. 84)

Man/machine interface and levels of automatic flight

PDF pages 86–91 · 18 questions

LEVEL 1 — RECALL

80. What is indicated by a green flight mode annunciation in the uppermost display line of a primary flight display ●

a. an armed operating mode
b. a failed operating mode
c. an active operating mode

81. At the managed automatic level of guidance, what supplies the reference values ●

a. the flight management system
b. the pilot at the control panel
c. the flight director channels

82. With which systems does the highest present level of automatic flight combine the flight management system ●

a. autopilot and the auto-throttle / auto-thrust system
b. flight director and the auto-throttle / auto-thrust system
c. autopilot and the flight director alone

LEVEL 2 — UNDERSTAND AND EXPLAIN

83. What does the operating mode selected on a mode control panel determine

a. the reference value or combination of values to be entered
b. the subsystem or combination of subsystems to be engaged
c. the regulator or combination of regulators to be activated

84. What do the regulations demand regarding the operating states or modes of the autoflight subsystems? ●

a. Feedback to the flight management system by a data bus
b. Feedback to the pilot by an aural warning tone
c. Feedback to the pilot by colourful annunciators

85. A pilot converting to another aircraft type cannot rely on the flight mode annunciator colours he is used to. Why not? ●

a. The colour of an annunciation depends on the reference value selected
b. There is no internationally agreed colour scheme for these annunciators
c. The colour scheme applies only to backlit panels in conventional cockpits

86. What are the reference values supplied by the flight management system at the managed automatic level based on ●

a. a stored flight plan and any modifications made to it in flight
b. a stored flight plan which cannot be modified once in flight
c. the air traffic control clearances passed during the flight

87. What is the objective of the managed automatic level of guidance ●

a. To let the crew leave the FMS profile at short notice
b. To support the pilot flying with attitude commands
c. To relieve the crew of manual control tasks

88. At the selected automatic level, what supplies the reference values ●

- **a.** the flight director from its selected modes
- **b.** the pilot at the autoflight control panel
- **c.** the flight management system from its flight plan

89. At the managed manual level, what supplies the reference values and parameters used by the FD ●

- **a.** the pilot at the control panel
- **b.** the flight management system
- **c.** the auto-thrust system

90. Where do the flight director inputs come from at the selected manual level ●

- **a.** supplied by the autopilot from its engaged basic modes
- **b.** manually set up on the autoflight control panel by the pilot
- **c.** supplied to the panel by the flight management system

91. At the basic manual level, what does the pilot flying have to do? ●

- **a.** Maintain altitude, heading and airspeed by selecting modes and reference values
- **b.** Maintain pitch and roll attitude by following the flight director commands
- **c.** Maintain altitude, heading and airspeed by manipulating the controls and thrust

92. At which level of guidance is the flight director off ●

- **a.** selected manual
- **b.** managed manual
- **c.** basic manual

LEVEL 3 — APPLY AND REASON

93. A blue annunciation appears in the second display line of a primary flight display. What does the pilot understand from it? ●

- **a.** The mode is armed and has been rejected by the autopilot
- **b.** The mode is active and is already controlling the aircraft
- **c.** The mode is armed and is waiting for an event to become active

94. The crew must leave the FMS profile at short notice after an air traffic control request, but wishes to keep the autopilot flying. Which level of guidance do they use? ●

- **a.** Selected manual
- **b.** Selected automatic
- **c.** Managed automatic

95. The autopilot is off, the flight director is on and FMS guidance is in use. Which level of guidance is this? ●

- **a.** Managed automatic
- **b.** Selected manual
- **c.** Managed manual

96. The autopilot and the auto-thrust system are on and there is no FMS guidance. Which level of guidance is this? ●

- **a.** Selected automatic
- **b.** Managed automatic
- **c.** Selected manual

97. At the three levels of guidance where the autopilot is off, in which states may the auto-throttle / auto-thrust system be ●

- **a.** off in every case
- **b.** on in every case
- **c.** on or off

Answers — Man/machine interface and levels of automatic flight

80. **c** Green on the top line means the mode is active. (L1, PDF p. 86)

81. **a** Managed means the FMS provides the reference values. (L1, PDF p. 90)

82. **a** The top level is the FMS working with the autopilot and the thrust system. (L1, PDF p. 90)

83. **c** Choosing a mode is choosing which regulators run. (L2, PDF p. 86)

84. **c** The rules call for the modes to be annunciated to the pilot in colour. (L2, PDF p. 86)

85. **b** There is no internationally agreed colour scheme for flight mode annunciators. (L2, PDF p. 86)

86. **a** The FMS works to the stored plan including whatever the pilot has changed in flight. (L2, PDF p. 90)

87. **c** Relieving the crew of the manual control tasks is the objective given, which leaves them time to monitor the overall situation. (L2, PDF p. 90)

88. **b** Selected means the pilot sets the reference values on the panel. (L2, PDF p. 90)

89. **b** Managed manual means the FD is fed by the FMS while the pilot flies by hand. (L2, PDF p. 90)

90. **b** In selected manual the pilot sets the FD modes and reference values himself. (L2, PDF p. 90)

91. **c** With no autoflight support the pilot holds the values himself on the controls and the thrust levers. (L2, PDF p. 90)

92. **c** Basic manual is the single row with the flight director off. (L2, PDF p. 91)

93. **c** Amber, white or blue on the second line is the armed indication, and armed means waiting for an event. (L3, PDF p. 86)

94. **b** Selected automatic leaves the autopilot in control while the pilot supplies the reference values himself. (L3, PDF p. 90)

95. **c** Autopilot off with flight director on and FMS guidance yes is the managed manual row. (L3, PDF p. 91)

96. **a** Autopilot on with no FMS guidance is the selected automatic row. (L3, PDF p. 91)

97. **c** Wherever the autopilot is off, the thrust system may be either on or off. (L3, PDF p. 91)

Flight condition control, engagement and interlocks

PDF pages 92–99 · 30 questions

LEVEL 1 — RECALL

98. Which flight condition parameters belong to the rotary degrees of freedom ●

a. pitch attitude, roll attitude and vertical speed
b. pitch attitude, roll attitude and yaw attitude
c. pitch rate, roll rate and load factor

99. What is the other name for yaw attitude ●

a. heading
b. drift
c. track

100. In which two states of operation is activation of an autopilot normally possible ●

a. CWS/MAN and CMD
b. CMD and APPROACH
c. CWS/MAN and NAV

101. Basic modes refer to flight condition parameters, so what do they consequently control ●

a. translatory degrees of freedom
b. rotary degrees of freedom
c. flight path parameters

102. Which connection is actually made when an autopilot is engaged ●

a. The torque limiter connecting the AP-servo with the trim actuator
b. The clutch connecting the AP-servo with the primary controls
c. The servo valve connecting the AP-servo with the hydraulic supply

103. Which device lets the pilot overpower an engaged autopilot at any time ●

a. A torque limiter
b. An engage interlock
c. An attitude command memory

104. What is the engage interlock circuit there to prevent ●

a. Activation of a defective or not yet synchronized autopilot
b. Activation of the manual trim while the autopilot is engaged
c. Activation of an upper mode before a basic mode is selected

LEVEL 2 — UNDERSTAND AND EXPLAIN

105. What do the flight condition regulators on their own make up ●

a. the simplest form of an autopilot
b. the mode control panel of an autopilot
c. the outer control loop of an autopilot

106. Whose control conditions are improved when flight condition regulators simplify and unify attitude control behaviour

a. the pilot or the outer control loops of the autopilot
b. the flight management system or the auto-thrust system
c. the pilot or the measurement unit of the control loop

107. Which change to the flying properties of an aircraft is brought about by its flight condition regulators? ●

a. Increased control forces at the control wheel
b. Increased effect of gusts on the aircraft
c. Increased damping of aircraft oscillations

108. Besides neutral stability, which modification of control properties is brought about by flight condition regulators ●

a. control wheel steering
b. control mode annunciation
c. control value monitoring

109. Besides preventing skid or slip in circling flight, what do the flight condition regulators provide ●

a. compensation of the wind correction angle
b. compensation of the thrust rating change
c. compensation of the loss of lift

110. The autopilot has been activated in CWS/MAN. What can the pilot do and in which modes does the autopilot work? ●

a. It works in the selected flight director modes and he must not touch the controls
b. It works in its upper modes and he can enter new reference values by hand
c. It works in its basic modes and he can influence the momentary attitude by hand

111. With the autopilot activated in CMD, which reference values does it use ●

a. Those stored in the flight management computer
b. Those set on the autoflight control panel
c. Those held from the moment of engagement

112. Besides the flight phase, what else do the names given to operating modes refer to ●

a. the reference value
b. the degree of freedom
c. the control value

113. Which regulator maintains the present heading in the basic mode heading hold ●

a. the yaw regulator
b. the roll regulator
c. the pitch regulator

114. The basic modes of some autopilots are heading hold together with vertical speed. What does the pitch regulator then maintain? ●

a. The present altitude
b. The present pitch attitude
c. The present climb rate

115. The pilot pushes firmly on the control column while the autopilot is engaged. What happens to the auto-trim function? ●

a. It is deactivated until the force is removed
b. It is transferred to the elevator actuator
c. It is switched over to its high trim speed

116. What decides whether disengagement follows an overpowering of the flight controls ●

a. The torque the limiter is built to pass and the airspeed
b. The circumstances of the input and the autopilot itself
c. The mode selected and the channel that is engaged

117. Every engage interlock condition is satisfied as the pilot selects the autopilot on. What follows? ●

a. The servo clutches open and the pilot keeps manual control
b. The servo clutches close and the autopilot takes over attitude control
c. The trim clutches close and the pilot keeps stabilizer control

118. What happens to the AP engage switch when an interlock condition becomes invalid ●

a. It is reset to the OFF position
b. It is latched in the ON position
c. It is held at its last selected mode

119. The engage interlock must establish that the autopilot is fit to take over. Which signals does it interrogate? ●

a. The signals held in the attitude command memory
b. The systems and signals of the outer flight path control loops
c. All required systems and signals of the inner control loop

LEVEL 3 — APPLY AND REASON

120. Why are flight condition regulators called basic regulators? ●

a. The outer control loops use them to achieve flight path control through attitude control
b. The outer control loops replace them whenever flight path control is required
c. The inner control loop uses them to achieve attitude control through flight path control

121. How are the problems of oscillation, gust effects and skid or slip actually solved? ●

a. By controlling and damping the rotary motions of the aircraft
b. By controlling and damping the translatory motions of the aircraft
c. By selecting an upper mode on the autoflight control panel

122. The pilot makes a control input that induces a roll rate or a load factor. What does the flight condition regulator do with it? ●

a. It converts the input into a reference value for the outer loop
b. It rejects the input until the autopilot has been disengaged
c. It transforms the input into the required dynamic behavior

123. Vertical speed is treated as a basic mode by some autopilots. Why is this called an exception? ●

a. Vertical speed is in fact a flight condition parameter
b. Vertical speed is in fact a flight path parameter
c. Vertical speed is in fact named after a flight phase

124. An autopilot in a basic mode is holding the attitude the aircraft had about its lateral axis. Which basic mode is in use? ●

a. Heading hold
b. Pitch attitude hold
c. Roll attitude hold

125. The aircraft is circling and the autopilot holds the bank it has. Which basic mode is this and which regulator carries it out? ●

- **a.** Heading hold, carried out by the yaw regulator
- **b.** Roll attitude hold, carried out by the roll regulator
- **c.** Roll attitude hold, carried out by the yaw regulator

126. The yaw damper drops out in flight while the autopilot is engaged. What is the consequence for the autopilot?

- **a.** An upper mode is lost, so the autopilot reverts to a basic mode
- **b.** A reference value is lost, so the autopilot re-synchronizes
- **c.** An interlock condition is lost, so the autopilot disengages

127. A crew member operates the manual trim while trying to engage the autopilot. Why will the autopilot not engage?

- **a.** An active manual trim input saturates the trim integrator
- **b.** An active manual trim input is an interlock condition not met
- **c.** An active manual trim input resets the attitude command memory

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Answers — Flight condition control, engagement and interlocks

98. b The three rotary parameters are pitch, roll and yaw attitude. (L1, PDF p. 92)

99. a Yaw attitude is also called heading. (L1, PDF p. 92)

100. a The two activation states are CWS/MAN and CMD. (L1, PDF p. 94)

101. b Flight condition parameters belong to the rotary freedoms, so basic modes control those. (L1, PDF p. 94)

102. b Engagement is the closing of the clutch between the AP-servo and the primary flight controls. (L1, PDF p. 96)

103. a The torque limiter is what permits forceful pilot inputs to override the autopilot. (L1, PDF p. 96)

104. a That is the stated purpose of the interlock circuit. (L1, PDF p. 98)

105. a An attitude stabilising regulator on its own is the simplest autopilot there is. (L2, PDF p. 92)

106. a Both the pilot and the outer loops of the autopilot get an easier aircraft to control. (L2, PDF p. 92)

107. c Better damping of the aircraft's oscillations is one of the listed modifications. (L2, PDF p. 92)

108. a Control wheel steering is named alongside neutral stability. (L2, PDF p. 92)

109. c The loss of lift in a turn is compensated as well as the skid or slip. (L2, PDF p. 92)

110. c In CWS/MAN the autopilot holds its basic modes while the pilot steers through it with the yoke. (L2, PDF p. 94)

111. b In CMD the references come from the autoflight control panel. (L2, PDF p. 94)

112. c A mode is named after either the control value it holds or the flight phase it serves. (L2, PDF p. 94)

113. b Heading hold is carried out by the roll regulator. (L2, PDF p. 94)

114. c In the vertical speed basic mode the pitch regulator holds the climb rate it has. (L2, PDF p. 94)

115. a Force on the control column deactivates auto-trim for as long as it is applied. (L2, PDF p. 96)

116. b Disengagement is made conditional on both the pilot's input and the design of the autopilot. (L2, PDF p. 96)

117. b Meeting all conditions closes the servo clutches and hands attitude control to the autopilot. (L2, PDF p. 98)

118. a Loss of a condition resets the engage switch to OFF. (L2, PDF p. 98)

119. c The interlock checks every required system and signal of the inner control loop before engagement is allowed. (L2, PDF p. 98)

120. a They are the building blocks the outer loops drive in order to steer the flight path. (L3, PDF p. 92)

121. a All of these are dealt with through control and damping of the rotary motions. (L3, PDF p. 92)

122. c Secondary parameters commanded by the pilot are turned by the regulator into the dynamic behaviour wanted. (L3, PDF p. 92)

123. b Basic modes normally hold flight condition parameters, and vertical speed is not one. (L3, PDF p. 94)

124. b The lateral axis is the pitch axis, so holding attitude about it is pitch attitude hold. (L3, PDF p. 94)

125. b Holding the present bank in a turn is roll attitude hold, done by the roll regulator. (L3, PDF p. 94)

126. c Yaw damper engagement is an interlock condition, and losing any condition disengages the servo clutches. (L3, PDF p. 99)

127. b The interlock requires that no manual trim be active, so the engage logic is blocked. (L3, PDF p. 99)

Control surfaces and dynamic response

PDF pages 100–102 · 11 questions

LEVEL 1 — RECALL

128. What supplies the force acting along the aircraft X-axis ●

a. Rudder deflection
b. Spoiler drag
c. Engine thrust

129. About which axis do the ailerons generate their moments ●

a. The longitudinal axis
b. The vertical axis
c. The lateral axis

LEVEL 2 — UNDERSTAND AND EXPLAIN

130. Thrust is counted as a primary control alongside the control surfaces. What singles it out from them? ●

a. It is the only control effective at low dynamic pressure
b. It is the only control that acts without a moment arm
c. It is the only way to supply energy to an aircraft

131. What job does the stabilizer do among the primary control surfaces ●

a. It trims out a yaw-moment generated by rudder deflection
b. It trims out a roll-moment generated by aileron deflection
c. It trims out a pitch-moment generated by elevator deflection

132. An engine failure leaves a thrust asymmetry in the cruise. Which primary control surface is held deflected to trim it out? ●

a. The stabilizer
b. The ailerons
c. The rudder

133. What is the difference in role between primary and secondary control surfaces ●

a. Primary surfaces give continuous control, secondary surfaces change the stationary condition
b. Primary surfaces give lateral control, secondary surfaces change the longitudinal condition
c. Primary surfaces give trimming control, secondary surfaces change the dynamic condition

134. Flaps and slats are extended for landing. In which direction is the flight envelope being extended? ●

a. Towards higher altitudes
b. Towards lower velocities
c. Towards higher velocities

135. An aircraft accelerates in level flight. Which quantity's rise makes its control surfaces more effective? ●

a. The dynamic pressure
b. The static pressure
c. The aircraft mass

LEVEL 3 — APPLY AND REASON

136. As the aircraft accelerates past a certain airspeed the maximum rudder deflection available is reduced. What is this limiting there to achieve? ●

a. It keeps the yaw damper inside its authority
b. It keeps the loads within the limits of structural strength
c. It keeps the surface within its maximum deflection angle

137. The ailerons are returned to neutral after a roll input. What does the aircraft then do?

a. It rolls back to a wings level attitude and stops circling
b. It keeps the roll attitude it has reached and circles
c. It keeps the roll rate it has reached and goes on rolling

138. Why does roll attitude show an integral relation to a constant aileron deflection

a. The bank angle stays constant because the roll rate decays to zero
b. The bank angle keeps increasing because the roll rate keeps rising
c. The bank angle keeps increasing because the roll rate stays constant

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Answers — Control surfaces and dynamic response

128. c Thrust acts along the X-axis. (L1, PDF p. 100)

129. a Aileron moments act about the longitudinal axis and are called roll moments. (L1, PDF p. 100)

130. c Thrust is treated as a primary control because it alone supplies energy to the aircraft. (L2, PDF p. 100)

131. c The stabilizer trims out the pitch moment the elevator has produced. (L2, PDF p. 100)

132. c A standing rudder deflection trims a thrust asymmetry. (L2, PDF p. 100)

133. a Primary surfaces are for continuous control; secondary surfaces change the stationary flight condition. (L2, PDF p. 100)

134. b Secondary surfaces extend the envelope towards lower speeds for take-off and landing. (L2, PDF p. 100)

135. a Control surface effectiveness is proportional to dynamic pressure, which rises with speed. (L2, PDF p. 100)

136. b Because effectiveness grows with dynamic pressure, deflection is limited at speed to hold loads within structural limits. (L3, PDF p. 100)

137. b Because roll response is integral, neutral ailerons leave the aircraft banked and turning. (L3, PDF p. 102)

138. c A steady roll rate accumulates with time into a steadily changing bank angle, which is integration. (L3, PDF p. 102)

Pitch modes, pitch trim, Mach and alpha trim

PDF pages 104–112 · 31 questions

LEVEL 1 — RECALL

139. Which pitch attitude does Pitch Attitude Hold maintain ●

a. The one held at the moment the mode is activated
b. The one scheduled for the present calibrated airspeed
c. The one held at the moment the flight director is selected

140. Which two phases are distinguished within the pitch regulator ●

a. Synchronization and control
b. Engagement and override
c. Synchronization and trimming

141. What is meant by a hard-over of a control surface ●

a. A deflection that reverses the commanded direction
b. A deflection held beyond the trim threshold
c. A full deflection through its whole travel range

142. What does the auto-trim function trim out while the autopilot is engaged ●

a. Short term pitch disturbances
b. Lateral trim asymmetries
c. Long lasting pitch disturbances

143. Which aspects are covered by the term flying qualities ●

a. Stability, controllability, trimmability and static margin
b. Controllability, maneuverability, damping and strength
c. Stability, controllability, maneuverability and gust ride

144. Which two pitch channel signals serve as the input to the auto-trim system ●

a. The attitude error or the elevator deflection
b. The pitch rate or the elevator deflection
c. The attitude error or the stabilizer position

145. What happens to the centre of lift as the Mach number increases ●

a. It moves aft
b. It stays fixed
c. It moves forward

146. Which unstable condition does the Mach trim subsystem exist to avoid ●

a. Tuck under
b. Dutch roll
c. Spiral divergence

LEVEL 2 — UNDERSTAND AND EXPLAIN

147. About which axis does Pitch Attitude Hold suppress disturbance moments ●

a. The longitudinal axis
b. The lateral axis
c. The vertical axis

148. What happens to the attitude command memory at the moment of engagement ●

a. Its input is restored and the momentary pitch angle is updated
b. Its output is interrupted and the momentary pitch angle is erased
c. Its input is interrupted and the momentary pitch angle is stored

149. The pitch error is scaled against the present calibrated airspeed before it reaches the actuator. Why is that done? ●

a. To obtain the same pitch attitude at all velocities
b. To obtain the same pitch rate at all velocities
c. To obtain the same elevator deflection at all velocities

150. Mach trim and speed trim are both fitted to an aircraft. Which flying quality do they improve? ●

a. The spiral stability
b. The gust ride qualities
c. The static stability

151. What do the alpha trim and alpha/Mach trim systems improve ●

a. The gust ride qualities
b. The maneuverability
c. The spiral stability

152. Which two safety demands does auto-trim fulfil ●

a. The elevator keeps its travel range and its actuator is unloaded in steady flight
b. The elevator keeps its travel range and its actuator is loaded in steady flight
c. The stabilizer keeps its travel range and its actuator is unloaded in steady flight

153. What is the effect of using a non-retroactive gear to drive the stabilizer ●

a. The stabilizer returns to neutral if the autopilot fails
b. The stabilizer stays in position if the autopilot fails
c. The stabilizer is driven by the elevator if the autopilot fails

154. In which two ways can the integration needed by the auto-trim system be obtained ●

a. By the trim actuator motor itself or by a distinct electronic integrator
b. By the stabilizer position sensor or by the trim actuator motor
c. By the trim detector itself or by a distinct electronic integrator

155. The trim speed is set differently for cruise and for approach. What makes that necessary? ●

a. The effectiveness of stabilizer and elevator depends on flap position
b. The travel range of stabilizer and elevator depends on airspeed
c. The effectiveness of stabilizer and elevator depends on airspeed

156. Which trim speeds are used for cruise and for approach ●

a. Low trim speed for cruise, low trim speed for approach
b. Low trim speed for cruise, high trim speed for approach
c. High trim speed for cruise, low trim speed for approach

157. What effect is produced by the movement of the centre of lift as the Mach number increases ●

a. A pitch down effect, which would increase the Mach number
b. A pitch down effect, which would decrease the Mach number
c. A pitch up effect, which would decrease the Mach number

158. High pressure differences at the wing tips are to be avoided in high speed flight. How does alpha trim achieve this? ●

a. By limiting the maximum stabilizer travel
b. By limiting the maximum Mach number
c. By limiting the maximum angle of attack

159. What does the alpha trim subsystem compare its sensed position against ●

a. A Mach trim function memory
b. An aircraft specific curve
c. A stabilizer trim unit scale

LEVEL 3 — APPLY AND REASON

160. A flight director mode is already active when the pilot engages the autopilot. How does the autopilot behave? ●

a. It holds the mode armed until the basic mode is cancelled
b. It takes the mode over and runs in the matching upper mode at once
c. It drops the mode and reverts to pitch attitude hold at once

161. Why is synchronization made one of the engage interlock conditions ●

a. It stops the actuators loading the trim system at engagement
b. It stops the actuators moving the surfaces abruptly at engagement
c. It stops the actuators reaching their travel limits in the cruise

162. During the synchronization phase the pitch regulator produces no actuator activity at all. Why is that so? ●

a. The momentary pitch angle is compared with itself, so the error is zero
b. The momentary pitch angle is held at zero, so the error is zero
c. The regulator output is clamped, so the actuator cannot respond

163. The pitch channel is said to behave like a PI-regulator. How is that behaviour produced? ●

a. The error goes direct to the stabilizer actuator and also to the auto-trim
b. The error goes direct to the elevator actuator and also to the gain schedule
c. The error goes direct to the elevator actuator and also to the auto-trim

164. An increase in thrust upsets the balance of pitch moments in the cruise. How is this disturbance classed and what does holding the attitude then need? ●

a. Short term, and a permanently generated additional moment
b. Permanent, and a momentary additional elevator deflection
c. Permanent, and a permanently generated additional moment

165. A lasting pitch disturbance has to be compensated with the autopilot engaged. Why is the stabilizer used rather than holding the elevator deflected? ●

a. The stabilizer must keep its full travel for long term disturbances
b. The elevator must keep its full travel for short term disturbances
c. The elevator must keep its neutral position for the gain schedule

166. A flight augmentation system fitted to the aircraft is unserviceable before departure. What is the likely effect on the autopilot?

a. Its availability is often a precondition for autopilot engagement
b. Its availability is only required for the upper modes of the autopilot
c. Its availability has no bearing because it works without the autopilot

167. The autopilot is disengaged after a long cruise during which auto-trim has been working. Why is there no abrupt pitch attitude change? ●

- **a.** The stabilizer is already resting in its neutral position
- **b.** The elevator is already resting in its neutral position
- **c.** The elevator is already held by the torque limiter

168. The Mach number rises in the cruise. Which way does the Mach trim drive the stabilizer and what does that counter? ●

- **a.** Leading edge down, countering the pitch down effect
- **b.** Leading edge down, countering the pitch up effect
- **c.** Leading edge up, countering the pitch down effect

169. The angle of attack climbs past the maximum allowed value. What does the alpha trim subsystem do and for how long? ●

- **a.** It drives the stabilizer leading edge up until alpha falls below the curve
- **b.** It drives the stabilizer leading edge down until alpha falls below the curve
- **c.** It drives the stabilizer leading edge up until the Mach number falls

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Answers — Pitch modes, pitch trim, Mach and alpha trim

139. a The reference is the attitude present when the mode becomes active. (L1, PDF p. 104)

140. a The two phases are synchronization and control. (L1, PDF p. 104)

141. c A hard-over is a run of the surface through its whole travel. (L1, PDF p. 106)

142. c Auto-trim deals with the lasting disturbances, not the short ones. (L1, PDF p. 108)

143. c Those four aspects are the ones listed. (L1, PDF p. 108)

144. a Either the attitude error or the proportional elevator deflection is used. (L1, PDF p. 110)

145. a The centre of lift travels rearwards with rising Mach number. (L1, PDF p. 112)

146. a The condition is named tuck under. (L1, PDF p. 112)

147. b Pitch moments act about the lateral axis. (L2, PDF p. 104)

148. c Engagement freezes the memory, so the stored pitch angle becomes the reference. (L2, PDF p. 106)

149. b The gain schedule keeps the pitch rate the same whatever the speed. (L2, PDF p. 106)

150. c Both are listed as static stability, that is longitudinal stability augmentation. (L2, PDF p. 108)

151. b These two are listed under maneuverability. (L2, PDF p. 108)

152. a Auto-trim keeps the elevator off its stops and keeps its actuator unloaded. (L2, PDF p. 110)

153. b The non-retroactive gear acts as a mechanical lock holding the stabilizer where it was left. (L2, PDF p. 110)

154. a Either the motor's own integral behaviour or a separate electronic integrator does it. (L2, PDF p. 110)

155. c Surface effectiveness varies with speed, so the trim rate is matched to it. (L2, PDF p. 110)

156. b Cruise uses the low trim speed, approach the high one. (L2, PDF p. 110)

157. a The nose drops and the aircraft accelerates further, which is why it is unstable. (L2, PDF p. 112)

158. c It caps the angle of attack. (L2, PDF p. 112)

159. b Sensed alpha is compared with the type's own curve. (L2, PDF p. 112)

160. b An active FD mode is adopted by the autopilot, so no basic mode appears. (L3, PDF p. 104)

161. b An unsynchronized autopilot would throw the surfaces over at engagement, so the interlock holds engagement off until synchronization is achieved. (L3, PDF p. 106)

162. a Both inputs to the summing point are the same signal, so the error is always zero. (L3, PDF p. 106)

163. c Splitting the error between the elevator actuator and auto-trim gives proportional plus integral action. (L3, PDF p. 106)

164. c A thrust change is a lasting disturbance, so the compensating moment must also be lasting. (L3, PDF p. 106)

165. b Holding the elevator deflected would eat into the travel it needs for gusts, so the stabilizer takes the standing load. (L3, PDF p. 106)

166. a These systems are commonly wired into the autopilot interlock. (L3, PDF p. 108)

167. b With the standing moment carried by the stabilizer and the elevator at neutral, nothing springs back at disengagement. (L3, PDF p. 110)

168. a Stabilizer leading edge down opposes the nose-down tuck under tendency. (L3, PDF p. 112)

169. a Leading edge up is nose down, which unloads the angle of attack until it is back under the curve. (L3, PDF p. 112)

Trim detector, roll modes, attitude and heading hold

PDF pages 114–124 · 28 questions

LEVEL 1 — RECALL

170. When is longitudinal stability considered to be present ●

a. When disturbances are counteracted by moments the autopilot commands at once
b. When disturbances are counteracted by moments the trimmed aircraft induces itself
c. When disturbances are counteracted by moments the stabilizer trim generates

171. Which control structure has one regulator supplying the reference value for a second regulator closer to the aircraft ●

a. Multi-loop or cascade control
b. Multi-rate or sampled control
c. Proportional or integral control

172. To which speed is the roll attitude adapted so that the rate of turn is held ●

a. The true airspeed
b. The indicated airspeed
c. The calibrated airspeed

173. Which three arrangements let the pilot keep control with the autopilot engaged ●

a. Turn or pitch knob control, control wheel steering and heading selector
b. Turn or pitch knob control, torque limiter and heading selector
c. Control wheel steering, heading selector and attitude command memory

174. On which autopilots is a heading selector always available ●

a. On all those with a heading indication at the mode controller
b. On all those with a heading select mode
c. On all those with a heading hold mode

LEVEL 2 — UNDERSTAND AND EXPLAIN

175. On which two factors does the trim threshold depend

a. Longitudinal static stability and flap position
b. Longitudinal static stability and dynamic pressure
c. Lateral static stability and dynamic pressure

176. What does the heading hold mode give the aircraft

a. Its missing positive spiral stability in roll
b. Its missing positive heading damping in yaw
c. Its missing positive static heading stability

177. Roll attitude hold is running with the autopilot engaged. What else is needed to improve the neutral spiral stability? ●

a. Turn compensation and a slight increase in engine thrust
b. Turn compensation and a slight reduction in engine thrust
c. Yaw damping and a slight increase in engine thrust

178. What are the roll attitude and heading command memories supplied with during synchronization ●

a. The reference roll and heading angles
b. The actual roll rate and heading error
c. The actual roll and heading angles

179. How is the roll attitude error signal adapted to the airspeed

a. By a gain schedule in the roll regulator
b. By a gain schedule in the attitude limiter
c. By a rate feedback in the roll regulator

180. What are control cascades primarily employed for ●

a. Simple processes whose control value cannot be measured
b. Complex processes made up of parallel independent processes
c. Complex processes made up of a sequence of sub-processes

181. What is done to the present bank angle before heading control commences

a. It is reduced to five degrees in a period of four seconds
b. It is reduced to zero in a period of ten seconds
c. It is reduced to zero in a period of four seconds

182. What does the roll outer loop put out ●

a. The reference heading for the heading inner loop
b. The aileron deflection for the roll servo loop
c. The reference roll attitude for the roll inner loop

183. What does turning a turn or pitch knob do ●

a. It modifies the actual attitude fed back to the inner loop
b. It modifies the reference value for the roll or pitch inner loop
c. It modifies the gain schedule for the roll or pitch inner loop

184. Where is turn and pitch knob control almost exclusively found ●

a. On newer control panels with no entry of reference values
b. On older control panels with no entry of reference values
c. On older control panels with no heading select mode

185. The CWS mode is selected with the autopilot engaged. How does the pilot then fly the aircraft? ●

a. Through the autopilot by means of the primary manual controls
b. Around the autopilot by means of the primary manual controls
c. Through the autopilot by means of the attitude reference knobs

186. Which arrangement is CWS equivalent to on a fly-by-wire aircraft ●

a. Turn and pitch knob control
b. Heading select mode
c. Side stick steering

LEVEL 3 — APPLY AND REASON

187. Fuel is burnt off through a long cruise and the centre of gravity moves aft. What happens to the aircraft's reaction to elevator and stabilizer displacements? ●

a. It changes, because the static stability has increased
b. It changes, because the static stability has decreased
c. It is unaltered, because the trim threshold is held fixed

188. An aircraft has a high longitudinal static stability. Why must the onset of auto-trim action be deferred? ●

a. Large stabilizer displacements are needed to counter long term disturbances
b. Large elevator displacements are needed to counter short term disturbances
c. Small elevator displacements are needed to counter short term disturbances

189. The elevator is oscillating slightly either side of the trim threshold. What stops the trim system becoming unstable?

a. Trimming is only allowed while the stabilizer position is changing
b. Trimming is only started at displacements above the drop out line
c. Trimming is only stopped at displacements below the drop out line

190. Why does an electronic integrator have to be added when an actuator motor is built into a servo loop ●

a. The motor loses the position feedback it needs for the loop
b. The motor loses its natural integral input/output behaviour
c. The motor loses its proportional input/output behaviour

191. An aircraft without the autopilot engaged is upset in roll by a gust. What is the lasting result once it has settled?

a. It is wings level again but on a different heading
b. It is wings level again and on its original heading
c. It holds the bank it reached and keeps turning

192. The autopilot is engaged while the aircraft is banked three degrees. What does the autopilot do? ●

a. It holds the three degrees of bank and lets the heading run
b. It rolls back to wings level and holds the heading of that moment
c. It rolls back to wings level and holds the last selected heading

193. The autopilot is engaged while the aircraft is banked ten degrees. What does the autopilot do? ●

a. It holds the bank angle, so the heading of that moment is held
b. It rolls back to wings level, so the heading of that moment is held
c. It holds the bank angle, so the heading changes and the aircraft turns

194. A regulator has to control a process made up of a long chain of sub-processes. Why does it need many feedback loops?

a. It requires feedback about the actual state of each sub-process
b. It requires feedback to keep each sub-process gain identical
c. It requires feedback about the real control value from each loop

195. An aileron deflection is held in. What does the airframe do with it before a heading change appears?

a. A roll rate is integrated into a bank angle, which gives lateral forces
b. A bank angle is integrated into a roll rate, which gives lateral forces
c. A roll rate is integrated into a yaw rate, which gives lateral forces

196. The aircraft is to hold the same rate of turn while flying faster through the air. What must happen to the bank angle? ●

a. It has to be held unchanged
b. It has to be increased
c. It has to be decreased

197. The autopilot has just been engaged and no upper mode has been selected. Which attitudes do its two channels control straight away? ●

a. The pitch channel the pitch attitude, the roll channel the roll attitude or heading
b. The pitch channel the flight path, the roll channel the roll attitude or heading
c. The pitch channel the pitch attitude, the roll channel the heading only

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Answers — Trim detector, roll modes, attitude and heading hold

170. b Stability means the aircraft itself produces the restoring moments. (L1, PDF p. 114)

171. a That arrangement is called multi-loop or cascade control. (L1, PDF p. 118)

172. a The heading outer loop gain is scheduled on true airspeed. (L1, PDF p. 122)

173. a Those three are the realizations listed. (L1, PDF p. 124)

174. b It is always there when the autopilot has heading select. (L1, PDF p. 124)

175. b Those are the two factors listed. (L2, PDF p. 114)

176. c Heading hold supplies the heading stability the airframe does not have. (L2, PDF p. 118)

177. a Turn compensation from the pitch channel plus a small thrust increase during circling do this. (L2, PDF p. 118)

178. c They track the actual angles until engagement freezes them. (L2, PDF p. 118)

179. a A gain schedule inside the roll regulator does the adaptation. (L2, PDF p. 120)

180. c They suit processes that are a chain of sub-processes. (L2, PDF p. 120)

181. c The wings are levelled over four seconds before heading control starts. (L2, PDF p. 122)

182. c Its output is the reference roll attitude handed down to the inner loop. (L2, PDF p. 122)

183. b The knob shifts the inner loop's reference value. (L2, PDF p. 124)

184. b It belongs to the older panels that had no way of entering reference values. (L2, PDF p. 124)

185. a The manual inputs are routed through the autopilot. (L2, PDF p. 124)

186. c CWS corresponds to side stick steering on fly-by-wire types. (L2, PDF p. 124)

187. b An aft CG lowers static stability, which alters how the aircraft answers elevator and stabilizer inputs. (L3, PDF p. 114)

188. b High stability makes the surfaces less effective, so ordinary gust corrections already use large deflections and trimming must wait. (L3, PDF p. 114)

189. c The drop out line sets a lower cut-off than the threshold, so small oscillations cannot chatter the system. (L3, PDF p. 114)

190. b Inside a servo loop the motor no longer integrates, so the behaviour must be restored electronically. (L3, PDF p. 116)

191. a Roll is self-recovering but heading is not, which is why heading hold is needed. (L3, PDF p. 118)

192. b Under five degrees the bank angle criterion selects heading hold on the heading present at engagement. (L3, PDF p. 118)

193. c At or above five degrees the bank is kept, which leaves the aircraft turning. (L3, PDF p. 118)

194. a Each sub-process has to be reported back, which is why there is a loop for each. (L3, PDF p. 120)

195. a The chain runs deflection to roll rate to bank angle to lateral force to heading change. (L3, PDF p. 120)

196. b Rate of turn at higher TAS needs more bank, which the outer loop gain schedule provides. (L3, PDF p. 122)

197. a Each channel takes its basic mode at once, pitch attitude in pitch and roll attitude or heading in roll. (L3, PDF p. 124)

Control wheel steering and manual modes

PDF pages 126–134 · 27 questions

LEVEL 1 — RECALL

198. How are the reference values generated when the autopilot is flown in CWS mode ●

a. By manual inputs on the control wheel or column
b. By preset inputs on the mode control panel display
c. By stored inputs from the flight management computer

199. Which devices at the flight controls sense a manual input by the pilot and turn it into an electrical signal ●

a. Force transducers, that is CWS sensors and dynamometric rods
b. Position pick-offs, that is synchros and resolver pick-ups
c. Rate sensors, that is tachometers and rate gyroscopes

200. What output is given by a force transducer or a dynamometric rod ●

a. A hydraulic pressure proportional to the applied force
b. An electrical current proportional to the applied travel
c. An electrical voltage proportional to the applied force

201. What does an ordinary autopilot engaged in command mode do when control forces exceed a certain level ●

a. It trims out the force
b. It disengages
c. It reverts to armed

202. At which break out force does the pitch channel switch off pitch attitude hold ●

a. More than 20 newtons
b. More than 50 newtons
c. More than 10 newtons

LEVEL 2 — UNDERSTAND AND EXPLAIN

203. Why do the manual controls end at positions equivalent to the pilot's own inputs during CWS operation ●

a. Because they are centred by springs inside the feel and centering unit
b. Because they follow up the movements of the control surfaces
c. Because they are locked by the autopilot engage clutch

204. What does the autopilot do if the roll angle on release of the yoke is less than 5 degrees ●

a. It keeps the present roll angle and holds the resulting turn rate
b. It decreases the roll angle to wings level and holds the previous heading
c. It decreases the roll angle to wings level and holds the present heading

205. What does the autopilot maintain when the yoke is released at a bank between 5 degrees and the type maximum ●

a. The present roll angle
b. The present turn rate
c. The present heading

206. What supports the improvement of neutral spiral stability in roll when CWS is used

a. Automatic turn compensation and possibly auto-thrust
b. Automatic lift compensation and possibly auto-brake
c. Automatic yaw damping and possibly auto-trim

207. What does an autopilot allowing supervisory override do when it senses force at the flight controls ●

a. It stays engaged and transits to CWS mode
b. It disengages and hands control back to the pilot
c. It stays engaged and transits to heading hold mode

208. When does an autopilot with supervisory override return to command mode ●

a. As soon as the pilot input is terminated, keeping its previously selected modes
b. Only after the pilot reselects command mode on the mode control panel
c. As soon as the pilot input is terminated, reverting to its basic attitude modes

209. What does the CWS sensor signal command during a pitch override input ●

a. An elevator deflection proportional to the force on the control column
b. A pitch attitude proportional to the force on the control column
c. A pitch rate proportional to the travel of the control column

210. What is held in the command memory of the pitch channel and what updates it during an override input ●

a. The elevator position reference, updated with the trim integrator output
b. The pitch rate reference, updated with the manually commanded rate
c. The pitch attitude reference, updated with the manually commanded attitude

211. What does the autopilot servo actuator do to the control column during a pitch override ●

a. It releases the column from the elevator so it can be moved freely
b. It tilts the column to a position matching the elevator deflection
c. It holds the column at the position the pilot has moved it to

212. Besides the roll attitude, which additional quantity must be synchronized during a roll override input ●

a. The aircraft turn rate
b. The aircraft heading
c. The aircraft slip angle

LEVEL 3 — APPLY AND REASON

213. A pilot holds a steady force on the control wheel in CWS. What do the inner loops make of it, and what does the aircraft do? ●

a. A fixed roll rate, which changes the heading but not the attitude
b. A proportional roll angle, which changes the turn rate and the track
c. A proportional roll rate, which changes the attitude and the heading

214. A pilot rolls the aircraft past its maximum roll angle in CWS and then releases the yoke. What does the autopilot do? ●

a. It holds the roll angle reached, beyond the maximum, until a new input
b. It rolls the aircraft back to wings level and maintains the heading reached
c. It rolls the aircraft back to the maximum roll angle and maintains it

215. A pilot flying in CWS makes a stick input and lets go. Why does the aircraft stay where he put it instead of going back?

- **a.** By turn compensation, so any stick input gives a constant pitch rate
- **b.** By artificial feel, so any stick input gives a rising stick force
- **c.** By auto-trim, so any stick input gives a new trimmed pitch attitude

216. Why do the computed control forces felt by a pilot flying in CWS stay the same as the centre of gravity, speed and weight change?

- **a.** Because the feel unit is rescheduled by the air data computer for them
- **b.** Because neutral stability makes the artificial feel independent of them
- **c.** Because the autopilot servo takes over the control loads created by them

217. A pilot applies force to the column and to the wheel at the same time in CWS. What two commands does the computer form from the sensor voltages? ●

- **a.** A pitch attitude or a roll rate command
- **b.** A vertical speed or a turn rate command
- **c.** A pitch rate or a roll attitude command

218. A pilot pushes the power levers forward while the thrust system is engaged. What do the rods fitted there sense and report? ●

- **a.** They sense the pilot input and signal a pilot override to the thrust controller
- **b.** They sense the lever travel and signal a thrust limit to the engine controller
- **c.** They sense the pilot input and signal a servo fault to the thrust controller

219. A pilot wants to nudge the aircraft while the autopilot is intercepting a localizer. What does supervisory override let him do? ●

- **a.** He can correct the autopilot during levelling off or an intercept without disengaging it
- **b.** He can recentre the controls during levelling off or an intercept without resynchronising them
- **c.** He can retrim the autopilot during levelling off or an intercept without reselecting it

220. A pilot is pressing on the controls of an autopilot that allows supervisory override. What is the state of the autopilot while he does so? ●

- **a.** The current modes are cancelled and the autopilot must be resynchronized
- **b.** The current modes are suspended but the autopilot stays synchronized
- **c.** The current modes are retained and the autopilot ignores the input

221. An aircraft in altitude hold is pushed into a climb by a pilot input at the column, and the column is then released. Which altitude does the autopilot now hold? ●

- **a.** The altitude selected before the control column was moved
- **b.** The altitude reached at the release of the control column
- **c.** The altitude midway between the two, blended by the trim integrator

222. The pilot reaches the speed he wanted and lets go of the control column after a supervisory override input. What does the pitch channel do at that moment? ●

- **a.** It waits for the trim integrator to run out before any mode is reactivated
- **b.** It sees the sensor signal fall to zero and reverts to the previous pitch attitude
- **c.** It sees the sensor signal fall to zero and reactivates pitch attitude hold at once

223. A CWS pitch input has left the elevator standing deflected to hold the new attitude. How is that lasting deflection dealt with? ●

- **a.** By trimming it out with the auto-trim system
- **b.** By cancelling the speed error with the auto-thrust system
- **c.** By masking it with the artificial feel unit

224. A pilot wants a different target airspeed while supervisory override is available and the autopilot is holding the nose where it is. How can he obtain it? ●

a. By changing the vertical speed target at the mode control panel
b. By changing the thrust setting manually at the power levers
c. By changing the pitch attitude manually at the control column

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Answers — Control wheel steering and manual modes

198. a In CWS the pilot's own hand input at the wheel or column is the reference value. (L1, PDF p. 126)

199. a The force measuring devices are force transducers, also called CWS sensors or dynamometric rods. (L1, PDF p. 126)

200. c Their output is a voltage that changes in proportion to the force applied at the input. (L1, PDF p. 128)

201. b Without supervisory override the autopilot simply drops out when the control force is high enough. (L1, PDF p. 134)

202. a The printed break out force is in excess of 20 newtons. (L1, PDF p. 134)

203. b The controls are tied to the surfaces, so they end up in positions matching the input made. (L2, PDF p. 126)

204. c Below 5 degrees of bank the autopilot rolls level and enters Heading Hold on the heading reached. (L2, PDF p. 126)

205. a Between 5 degrees and the maximum the bank angle present at release is the value held. (L2, PDF p. 126)

206. a Turn compensation, with auto-thrust where fitted, is what supports the improved spiral stability. (L2, PDF p. 126)

207. a Supervisory override converts a force on the controls into a change to CWS rather than a disengagement. (L2, PDF p. 134)

208. a Ending the manual input is enough for the autopilot to go back to command with its earlier modes. (L2, PDF p. 134)

209. b The commanded quantity is a pitch attitude that follows the applied column force. (L2, PDF p. 134)

210. c The memory stores the pitch attitude reference and takes on the value the pilot commands by hand. (L2, PDF p. 134)

211. b The servo extends or retracts so the column sits where the elevator deflection corresponds. (L2, PDF p. 134)

212. b In roll both the roll attitude and the heading have to be kept synchronized. (L2, PDF p. 134)

213. c The force becomes a proportional roll rate, and a lasting roll rate changes the attitude and so the heading. (L3, PDF p. 126)

214. c Beyond the limit the autopilot recovers to the maximum roll angle and holds it in Roll Attitude Hold. (L3, PDF p. 126)

215. c Neutral pitch stability is built in, and auto-trim retrims the aircraft at whatever attitude the input produced. (L3, PDF p. 126)

216. b Neutral stability is the reason the artificial feel forces do not change with CG, speed or weight. (L3, PDF p. 126)

217. a The computer makes a pitch attitude command or a roll rate command out of the sensor voltage. (L3, PDF p. 128)

218. a The rods pick up the hand force on the levers and tell the thrust controller that the pilot is overriding. (L3, PDF p. 128)

219. a The point of supervisory override is that the pilot can help the autopilot without dropping it out. (L3, PDF p. 134)

220. b Modes are only suspended during the input, and the autopilot remains synchronized throughout. (L3, PDF p. 134)

221. b The value reached when the pilot lets go becomes the new reference for the suspended upper mode. (L3, PDF p. 134)

222. c Zero sensor signal is the trigger that immediately brings pitch attitude hold back. (L3, PDF p. 134)

223. a Auto-trim takes out the standing elevator deflection so the new attitude is held. (L3, PDF p. 134)

224. c With pitch attitude hold active, a manual pitch change is the way the pilot trades to a new speed. (L3, PDF p. 134)

Heading select and cross coupling

PDF pages 136–142 · 24 questions

LEVEL 1 — RECALL

225. What does the heading indication at the mode controller show when power is first applied on the ground ●

a. Either the last selected heading or the magnetic variation
b. Either the present heading or the last selected heading
c. Either the present heading or the runway heading

226. Which heading do some autopilots display at their mode control panel when power is first applied on the ground ●

a. 000 degrees
b. 180 degrees
c. 360 degrees

227. How many control loops make up the heading select control cascade ●

a. One
b. Three
c. Two

228. What does the term cross coupling mean ●

a. The mutual influence of the aircraft axes when attitude or thrust is changed
b. The mutual influence of the control surfaces when a servo channel is engaged
c. The mutual influence of the control loops when a mode is reselected

229. Under which conditions is a horizontal turn said to be coordinated ●

a. The aircraft neither slips nor skids and holds its altitude and airspeed
b. The aircraft neither climbs nor descends and holds its bank and thrust
c. The aircraft neither slips nor skids and holds its bank and turn rate

LEVEL 2 — UNDERSTAND AND EXPLAIN

230. What does the autopilot do once the heading select mode is engaged ●

a. It arms the mode until the pre-selected heading is reached
b. It turns the aircraft to the pre-selected heading and maintains it
c. It turns the aircraft onto the present heading and then maintains it

231. The crew has a heading mode running and air traffic control passes a different heading. When may the selected heading be altered? ●

a. At any time while the mode is active
b. Only after the first heading has been captured
c. Only after the mode has been reselected

232. A pilot selects the heading mode on the mode control panel. Which loop of the cascade is closed by that selection? ●

a. The outer loop
b. Both loops together
c. The inner loop

233. How is the actual value of the heading obtained for the outer loop ●

a. From a compass system or an inertial reference system
b. From a directional gyro or a radio magnetic indicator
c. From a compass system or an air data computer

234. Where and how is the heading error formed ●

a. At S1 in the outer loop, by comparing actual with commanded roll angle
b. At S1 in the outer loop, by comparing actual with reference heading
c. At S2 in the inner loop, by comparing actual with reference heading

235. An aircraft shows cross coupling between its axes. Why must those effects be kept as small as possible?

a. They increase the control forces felt at the control column
b. They reduce the authority of the autopilot servo actuators
c. They stimulate motion in the affected degrees of freedom

236. What does the design of a modern autoflight system ensure about its own cross couplings ●

a. They counteract the cross coupling effects in the aircraft
b. They reproduce the cross coupling effects in the aircraft
c. They are removed entirely from the control channels

237. A turn is about to be started. What generates the rolling moment that begins it? ●

a. Deflection of the rudder
b. Deflection of the elevator
c. Deflection of the ailerons

238. What forces the aircraft onto a circular flight path once it is banked ●

a. The horizontal component of the total drag vector
b. The horizontal component of the tilted lift vector
c. The vertical component of the tilted lift vector

239. What is the single cause of both the lateral accelerations and the unfavourable aerodynamic effects in a turn ●

a. The existence of a pitch error
b. The existence of a slip angle
c. The existence of a roll rate

240. An aircraft is turning with a slip angle present. How is that slip angle brought to zero? ●

a. By deflecting the rudder in coordination with the ailerons
b. By deflecting the elevator in coordination with the ailerons
c. By deflecting the spoilers in coordination with the rudder

LEVEL 3 — APPLY AND REASON

241. A meal is being served in the cabin and the pilot winds the outer knob below the heading window to its lower setting. What is he achieving? ●

a. A smaller maximum roll angle, which improves passenger comfort
b. A smaller maximum heading change, which improves passenger comfort
c. A slower heading capture rate, which improves passenger comfort

242. What is the heading error turned into before it leaves the outer control loop ●

- **a.** A roll rate reference value fed to the inner loop
- **b.** A roll angle reference value fed to the inner loop
- **c.** An aileron position demand fed to the servo

243. The reference value reaching the inner loop no longer matches the actual value fed back to it. What does the inner loop do? ●

- **a.** It commands a change in heading reference
- **b.** It commands a change in rudder deflection
- **c.** It commands a change in roll attitude

244. The autopilot banks the aircraft into a turn and must not let it descend. Why does the elevator have to be deflected as well? ●

- **a.** Total lift must fall so its horizontal part still curves the path
- **b.** Total lift must rise so its vertical part still balances the weight
- **c.** Total drag must rise so the airspeed stays at the reference value

245. An altitude or vertical speed controller is left to correct a turn on its own, with no lift compensation fitted. What is the consequence? ●

- **a.** It reacts too early and causes a lasting offset from the reference value
- **b.** It reacts too late and causes large errors or even unstable pitch control
- **c.** It reacts correctly but leaves a permanent slip angle in the turn

246. A turn is entered and the bank angle goes on increasing. How does the input the pitch channel receives from the lateral channel change? ●

- **a.** It increases aileron deflection in proportion to the existing pitch angle
- **b.** It increases elevator deflection in proportion to the existing turn rate
- **c.** It increases elevator deflection in proportion to the existing roll angle

247. The autopilot increases lift to hold altitude in a turn. Why must the thrust system then be brought in as well? ●

- **a.** Because the extra lift brings extra weight, which would lower the ceiling
- **b.** Because the extra bank brings extra slip, which would raise the fuel flow
- **c.** Because the extra lift brings extra drag, which would lower the airspeed

248. The autoflight system lets the altitude wander slightly instead of correcting it at once. What does the airline gain from that?

- **a.** Fewer thrust changes, so less fuel is burnt and comfort improves
- **b.** Fewer altitude errors, so less drag is created and comfort improves
- **c.** Fewer mode changes, so less wear occurs and comfort improves

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Answers — Heading select and cross coupling

225. b At power up the display is either synchronized to the present heading or shows the last selection. (L1, PDF p. 138)

226. a Some autopilots display 000 in this condition. (L1, PDF p. 138)

227. c The cascade is made up of two control loops. (L1, PDF p. 140)

228. a Cross coupling is defined as one axis influencing another when attitude or thrust rating is changed. (L1, PDF p. 142)

229. a A coordinated horizontal turn is free of slip and skid and keeps altitude and airspeed. (L1, PDF p. 142)

230. b Selecting the mode makes the autopilot fly to the pre-selected heading and then hold it. (L2, PDF p. 136)

231. a The selected heading may be changed at any time while heading select is running. (L2, PDF p. 136)

232. a Selecting the mode closes the outer loop of the cascade. (L2, PDF p. 140)

233. a The actual heading comes from a compass system or from the IRS. (L2, PDF p. 140)

234. b S1 in the outer loop is where actual and reference heading are compared. (L2, PDF p. 140)

235. c They are undesirable because they set up motion in the degrees of freedom they affect. (L2, PDF p. 142)

236. a The couplings built into the system are there to work against the ones the airframe produces. (L2, PDF p. 142)

237. c The ailerons are deflected to produce the rolling moment that starts the turn. (L2, PDF p. 142)

238. b Tilting the lift vector gives a horizontal component, and that is what curves the flight path. (L2, PDF p. 142)

239. b Both arise from a slip angle being present. (L2, PDF p. 142)

240. a Rudder deflection coordinated with the aileron deflection takes the slip out. (L2, PDF p. 142)

241. a Reducing the maximum roll angle is what that knob does, and comfort is the stated reason for using it. (L3, PDF p. 138)

242. b The outer loop converts heading error into a roll angle reference for the inner loop. (L3, PDF p. 140)

243. c A difference between reference and actual roll angle makes the inner loop command a roll attitude change. (L3, PDF p. 140)

244. b Banking tilts the lift, so total lift must be increased to keep the vertical part equal to the weight. (L3, PDF p. 142)

245. b Altitude and vertical speed change slowly, so the controller is late and large errors or instability follow. (L3, PDF p. 142)

246. c Lift compensation adds elevator in proportion to the bank angle to raise the vertical lift component. (L3, PDF p. 142)

247. c More lift means more drag and a loss of speed, so a roll to thrust coupling adds thrust to hold the speed. (L3, PDF p. 142)

248. a Trading energy cuts the number of thrust changes, which saves fuel and makes the ride smoother. (L3, PDF p. 142)

Flight path guidance and level change mode

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LEVEL 1 — RECALL

249. From which two sources does the airspeed or Mach reference number come ●

a. Dialled by the pilot at the mode control panel or supplied by the ADC
b. Dialled by the pilot at the mode control panel or supplied by the FMS
c. Computed by the autopilot from the flight plan or supplied by the FMS

250. What is the purpose of the LVL/CH mode ●

a. To transfer speed control automatically to the auto-thrust system
b. To transfer the aircraft automatically to a higher or lower flight level
c. To transfer the aircraft automatically onto a selected vertical speed

251. Which computer normally calculates the thrust limits for each flight phase ●

a. The flight control computer
b. The flight warning computer
c. The FADEC computer

252. In which memory is the reference value stored ●

a. The speed and Mach memory
b. The altitude capture memory
c. The attitude command memory

253. What is the source of the control value used for airspeed or Mach number control ●

a. The air data computer
b. The inertial reference system
c. The flight management system

LEVEL 2 — UNDERSTAND AND EXPLAIN

254. What part does the pilot himself play when he is flying the aircraft along its flight path by hand ●

a. He makes up a feedback path for the attitude command memory
b. He makes up an outer loop for the translatory degrees of freedom
c. He makes up an inner loop for the three rotational degrees of freedom

255. How did the term autopilot become established

a. From the automatic trim function of relieving the loads at the control column
b. From the automatic stability function of damping the aircraft about its axes
c. From the automatic pilot function of guiding the aircraft along its flight path

256. A flight path controller is doing its job on a normal sector. Which upper autopilot modes do its tasks correspond to? ●

a. Takeoff, climb, descend, flare and automatic braking
b. Climb, cruise, hold, go around and automatic taxi
c. Climb, enroute, descend, approach and automatic landing

257. What does a flight management system relieve the pilot of ●

a. Most manual control tasks, which it trims automatically
b. Most monitoring tasks, which it displays automatically
c. Most routine tasks, which it sequences automatically

258. An FMS is sequencing a flight path segment by segment. What does it do automatically at each one? ●

a. Choosing navigation aids and tuning their frequencies for each path segment
b. Choosing engine ratings and setting their thrust limits for each path segment
c. Choosing autopilot channels and testing their servos for each path segment

259. Which quantities does a flight management system calculate for lowest fuel consumption ●

a. Optimal thrust limits, bleed air settings and flap schedules
b. Optimal bank angles, turn radii and track changes
c. Optimal climb and sink rates, cruise altitudes and airspeeds

260. How does the autopilot pitch channel control airspeed or Mach number ●

a. By altering the thrust rating through the power lever position
b. By altering the drag through the speed brake panel setting
c. By altering the flight path angle through the aircraft's attitude

261. How widely is the airspeed or Mach reference value applied once it is set ●

a. It is always applied to both speed control systems
b. It is always applied to the system holding priority only
c. It is always applied to the auto-thrust system only

262. Under which condition does the autoflight system maintain the reference airspeed or Mach number during a level change ●

a. As long as no autopilot or auto-thrust channel has failed
b. As long as no thrust or flight envelope limits are exceeded
c. As long as no altitude or vertical speed limits are exceeded

263. What does thrust alone have to overcome during an unaccelerated horizontal flight ●

a. Drag
b. Inertia
c. Weight

264. Which system takes charge of airspeed or Mach number the moment a level change is triggered ●

a. The autopilot, which holds it by adjusting the flight path angle
b. The autopilot, which holds it by adjusting the thrust rating
c. The auto-thrust, which holds it by adjusting the thrust rating

265. The crew select a higher flight level and trigger a level change. What thrust does the auto-thrust system set? ●

a. The maximum thrust for the present flight phase
b. The flight idle thrust for the present flight phase
c. The cruise thrust for the present flight phase

266. What does the auto-thrust system do for a level change in the descending direction ●

a. It sets the flight idle rating
b. It disengages until capture
c. It sets the maximum continuous rating

267. What do the thrust limits depend on ●

a. Actual gross weight and the amount of bleed air tapped from an engine
b. Actual air data and the amount of fuel remaining in the tanks
c. Actual air data and the amount of bleed air tapped from an engine

268. Why is the capture point needed ●

a. To give a smooth transition from a climb or descent to level flight
b. To give an accurate indication of the time left to the target level
c. To give a smooth transition from thrust mode to speed mode

269. The aircraft is in the capture phase and the altitude error is shrinking. How is the vertical speed brought down? ●

a. By decreasing the thrust rating in proportion to the decreasing altitude error
b. By decreasing the flight path angle in proportion to the decreasing speed error
c. By decreasing the flight path angle in proportion to the decreasing altitude error

270. What happens to speed control when altitude acquisition becomes active ●

a. The pitch channel hands it to the thrust system, which goes to speed mode
b. The thrust system hands it to the pitch channel, which goes to speed mode
c. The pitch channel keeps it and the thrust system goes to thrust mode

271. Which is the only system performing speed control in IAS hold or Mach hold ●

a. The autothrottle system
b. The flight management system
c. The autopilot pitch channel

272. An autopilot offers only IAS hold and Mach hold. What is the limitation of that control mode? ●

a. It does not allow a change over between speed and Mach
b. It does not allow bracketing of a pre-selected reference speed
c. It does not allow the autothrottle to reach its thrust limit

273. Which two names are used for the reference value according to where it comes from ●

a. Selected speed target or computed speed target
b. Preset speed target or managed speed target
c. Selected speed target or managed speed target

274. An aircraft climbs to the level at which speed control passes from airspeed to Mach number. What is that altitude called? ●

a. The optimum or cruise altitude
b. The capture or transition altitude
c. The crossover or changeover altitude

275. How does the autoflight system stop an entered speed target from exceeding the flight envelope

a. By warning the crew and leaving the entered value in force
b. By rejecting the entered value and reverting to the last valid one
c. By voting between the entered value and the computed limits

276. At which pressure altitude is the automatic change over performed during a climb or a descent ●

a. Approximately 18 000 feet
b. Approximately 36 000 feet
c. Approximately 26 000 feet

277. An aircraft climbs through the changeover altitude and the reference changes over by itself. Which system has done it? ●

a. The auto-throttle and auto-thrust system
b. The air data and inertial reference system
c. The autopilot and flight director system

278. How is the airspeed and Mach control loop built up ●

a. An outer loop is added to the roll attitude controller
b. An outer loop is added to the pitch attitude controller
c. An inner loop is added to the altitude hold controller

279. Which autoflight mode sets up the airspeed and Mach outer control loop when it is selected

a. Heading select
b. Altitude hold
c. Level change

280. The speed controller is running in level change. Where is the speed error formed, and from what? ●

a. At S2, by comparing reference speed with actual speed
b. At S2, by comparing reference attitude with actual speed
c. At S1, by comparing reference attitude with actual attitude

281. What are the two jobs of the pitch attitude limiter ●

a. It curtails extreme commands and sets the gain schedule of the inner loop
b. It curtails extreme commands and sets the timing and direction of the change
c. It curtails the speed error and sets the timing and direction of the change

LEVEL 3 — APPLY AND REASON

282. Gusts push an aircraft away from its altitude and its track. Why can the inner control loops not put that right on their own?

a. The inner loops cannot compensate disturbances acting on the attitude
b. The inner loops cannot compensate disturbances acting on the flight path
c. The inner loops cannot drive the control surfaces without a servo

283. Leaving aside the thrust of the engines, how is the flight path of an aircraft changed in principle ●

a. By acting on the rotary degrees of freedom, that is by changing attitude
b. By acting on the thrust setting, that is by changing engine rating
c. By acting on the aerodynamic configuration, that is by changing flap setting

284. Why does the autopilot have to treat interception and stabilization as two separate jobs? ●

a. The path is in segments, and each must be caught without overshoot or offset
b. The path is continuous, but the sensors update at two different rates
c. The path is in segments, and each one has to be flown at a different airspeed

285. What must be in place before an FMS can guide an aircraft laterally and vertically from departure to destination?

a. A flight plan of waypoints and the relevant modes selected for the autoflight system
b. A flight plan of waypoints and the relevant frequencies tuned at the radio panel
c. A route clearance entered and the relevant altitudes set at the mode control panel

286. Both the pitch channel and the thrust system can hold the speed. Which of them gets the job, and why can they not share it? ●

a. The autopilot, because the flight management system cannot supply two targets
b. The autopilot, because the two systems cannot both control speed at once
c. The auto-thrust, because the two systems cannot both control speed at once

287. The autopilot pitch channel lowers the nose while it is controlling speed. What energy change takes place and what does the speed do? ●

a. Potential energy becomes kinetic energy and the speed decreases
b. Kinetic energy becomes potential energy and the speed decreases
c. Potential energy becomes kinetic energy and the speed increases

288. The pilot selects altitude hold, so the pitch channel is now given another job. Who controls the speed and how? ●

a. The auto-thrust system, by thrust control to a reference speed
b. The flight management system, by thrust control to a reference speed
c. The auto-thrust system, by drag control to a reference speed

289. The autopilot pitch channel is taken off speed control and put onto altitude hold. What does that decide for the thrust system? ●

a. Its engagement, which the pitch channel determines
b. Its mode of operation, which the pitch channel determines
c. Its thrust limit, which the pitch channel then determines

290. The autopilot is controlling speed in level flight and the thrust is then increased. What does the autopilot do with the aircraft? ●

a. It increases the flight path angle and the aircraft begins to accelerate
b. It increases the flight path angle and the aircraft begins to climb
c. It decreases the flight path angle and the aircraft begins to climb

291. A level change climb is being flown. What decides the amount of pitch change and the climb rate that result? ●

a. Only the pitch attitude limit applied by the autopilot pitch channel
b. Only the surplus of thrust from the rating set by the thrust system
c. Only the altitude error between present and preselected flight level

292. How does the thrust system decide whether to apply its thrust limit or flight idle at the start of a level change? ●

a. It evaluates the size of the altitude error, taken as preselected minus actual
b. It evaluates the sign of the speed error, taken as reference minus actual
c. It evaluates the sign of the altitude error, taken as preselected minus actual

293. An aircraft is climbing fast towards the selected level. How does its high vertical speed affect the point at which the capture starts? ●

a. The capture point is unaffected, being a fixed distance from the level
b. The capture point is placed closer to the level, in proportion to it
c. The capture point is placed further from the level, in proportion to it

294. The aircraft reaches the capture point during a level change climb. Which mode changes take place? ●

a. Level change is rearmed and altitude acquisition is turned off
b. Level change is turned off and altitude acquisition is activated
c. Level change is turned off and altitude hold is activated

295. Why can the altitude acquisition law not simply be left running until the selected level is reached? ●

a. Its transition follows a ramp function, which would overshoot the level
b. Its transition follows an e-function, which would take an infinite time
c. Its gain is scheduled with airspeed, which would change during the capture

296. When should a crew with only IAS hold available select it during a climb or a descent? ●

a. Once the new flight level has been selected
b. Once the new reference speed has been reached
c. Before the thrust system reaches its climb limit

297. An aircraft climbs through the changeover altitude with a speed target that was dialled in by hand. What becomes of the number in the window? ●

a. It is replaced by dashes until the pilot enters a Mach number
b. It is converted to the equivalent true airspeed and displayed there
c. It is converted to the equivalent Mach number and displayed there

298. The crew look at the speed window on the mode control panel and see dashes. What does that tell them? ●

a. The speed target is outside the limits of the flight envelope
b. The speed target has been lost and the autothrottle has disengaged
c. The speed target is being supplied by the flight management system

299. A Mach target is being flown at a high flight level. What does the reference speed marker on the primary flight display show? ●

a. A calibrated airspeed, because the tape is in CAS at any altitude
b. A Mach number, because the tape follows the active reference
c. An indicated airspeed, because the tape is in IAS at any altitude

300. The actual speed differs from the stored reference when a speed mode is armed. Why is the stored value synchronized first, and how? ●

a. To avoid a hard over, by storing the envelope limit at mode activation
b. To avoid a hard over, by storing the actual value at mode activation
c. To avoid an envelope exceedance, by storing the actual value at mode activation

301. The flaps and the landing gear are extended. Which of the two configuration speed limits does VMAX take? ●

a. The lower of VFE and VLE
b. The higher of VFE and VLE
c. The lower of VMO and MMO

302. A crew wants to work to a Mach reference well below the normal transition height. What can they do and where? ●

a. Re-enter the target as a Mach number in the flight management system
b. Press the crossover pushbutton beside the altitude select display
c. Press the crossover pushbutton beside the speed and Mach display

303. Which parameter is held constant above the automatic change over level and which below

a. Indicated airspeed above and Mach number below
b. Mach number above and indicated airspeed below
c. True airspeed above and indicated airspeed below

304. What does the PI behaviour of the outer loop prevent, and how ●

a. A standing speed error, by limiting the pitch attitude being commanded
b. A standing speed error, by integrating it into the attitude reference
c. An unstable speed loop, by integrating it into the thrust demand

305. A descent at a lower reference speed is commanded. In what order does the pitch controller act? ●

a. It reduces the speed error and pitches down at the same time
b. It reduces the speed error to zero first and then pitches down
c. It pitches down first and then reduces the speed error to zero

306. What is the result of changing pitch attitude whenever a speed error appears at a given thrust

a. The aircraft adopts a thrust rating that restores the reference speed
b. The aircraft adopts a flight path angle that restores the reference speed
c. The aircraft adopts a vertical speed that restores the reference altitude

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Answers — Flight path guidance and level change mode

249. b It is either dialled by hand on the mode control panel or passed over by the FMS. (L1, PDF p. 145)

250. b The mode exists to take the aircraft automatically from one flight level to another. (L1, PDF p. 146)

251. c The FADEC computer works out the thrust limits for each flight phase. (L1, PDF p. 146)

252. a It is held in the speed and Mach memory. (L1, PDF p. 150)

253. a Calibrated airspeed or Mach comes from the air data computer. (L1, PDF p. 154)

254. b A hand-flying pilot is himself the outer loop for the translatory degrees of freedom. (L2, PDF p. 144)

255. c The name comes from the system doing the pilot's job of guiding the aircraft along the flight path. (L2, PDF p. 144)

256. c Those five are the upper modes listed. (L2, PDF p. 144)

257. c The FMS takes over most routine tasks and sequences the events itself. (L2, PDF p. 144)

258. a Selecting the right navigation aids and tuning them is the example given. (L2, PDF p. 144)

259. c Those are the optimal values the FMS works out for the lowest fuel burn. (L2, PDF p. 144)

260. c The pitch channel changes the flight path angle by changing pitch attitude with the elevator. (L2, PDF p. 145)

261. a The one reference value goes to both of the speed controlling systems. (L2, PDF p. 145)

262. b The reference speed is held while thrust limits and envelope limits are respected. (L2, PDF p. 146)

263. a In level unaccelerated flight the thrust only has to balance the drag. (L2, PDF p. 146)

264. a At level change the autopilot always takes the speed and holds it with flight path angle. (L2, PDF p. 146)

265. a For a climb the system sets the maximum thrust applicable to the flight phase. (L2, PDF p. 146)

266. a A descent needs a lack of thrust, so flight idle is set. (L2, PDF p. 146)

267. c They depend on the air data of the moment and on how much bleed air is being taken. (L2, PDF p. 146)

268. a It exists so that the levelling off at the selected flight level is smooth. (L2, PDF p. 146)

269. c The flight path angle is reduced in step with the shrinking altitude error. (L2, PDF p. 146)

270. a The pitch channel is now needed for the capture, so speed passes to the thrust system in speed mode. (L2, PDF p. 146)

271. a In this mode the autothrottle alone does the speed control. (L2, PDF p. 148)

272. b There is no bracketing of a pre-selected reference speed in this mode. (L2, PDF p. 148)

273. c The two printed names are selected speed target and managed speed target. (L2, PDF p. 150)

274. c That altitude is called the cross or changeover altitude. (L2, PDF p. 150)

275. c The system votes between the entered value and the limits so the envelope is not exceeded. (L2, PDF p. 150)

276. c The printed altitude is approximately 26 000 feet. (L2, PDF p. 152)

277. c The AP/FD system carries out the automatic change over. (L2, PDF p. 152)

278. b The pitch attitude controller gains an outer loop to make the speed controller. (L2, PDF p. 154)

279. c Selecting level change is what makes the mode logic build the outer loop. (L2, PDF p. 154)

280. a S2 is the summing point where reference and actual speed are compared. (L2, PDF p. 154)

281. b It both limits extreme attitude commands and governs when and which way the pitch change happens. (L2, PDF p. 154)

282. b Inner loops stabilize attitude only, so outer loops are added for flight path disturbances. (L3, PDF p. 144)

283. a The aircraft is moved about its path by changing its attitude. (L3, PDF p. 144)

284. a Because the path is built of segments, each one has to be intercepted cleanly and then held. (L3, PDF p. 144)

285. a Both a waypoint flight plan and the FMS modes selected for the autoflight system are required. (L3, PDF p. 144)

286. b Only one may control speed, and the priority is given to the autopilot. (L3, PDF p. 145)

287. c Pitching down converts height into speed, so the aircraft accelerates. (L3, PDF p. 145)

288. a With the pitch channel busy, speed passes to the auto-thrust, which holds it with thrust. (L3, PDF p. 145)

289. b In current systems what the pitch channel is doing sets the mode the thrust system runs in. (L3, PDF p. 145)

290. b The autopilot holds speed by raising the flight path angle, so the surplus thrust becomes a climb. (L3, PDF p. 146)

291. b Both depend only on the surplus of thrust left after the thrust system has set its rating. (L3, PDF p. 146)

292. c The sign of the altitude error tells the system whether the aircraft must go up or down. (L3, PDF p. 146)

293. c The capture point is a vertical distance from the selected level set in proportion to the vertical speed of the moment. (L3, PDF p. 146)

294. b At the capture point level change drops out and altitude acquisition takes over. (L3, PDF p. 146)

295. b The capture is exponential, so in theory it never quite arrives and must be terminated. (L3, PDF p. 146)

296. b Because it only freezes the current value, it is selected after the wanted speed has been reached. (L3, PDF p. 148)

297. c At the transition the value is converted to the equivalent Mach number, and a value entered at the panel is shown there. (L3, PDF p. 150)

298. c Dashes in the window mean the target is managed by the FMS rather than selected by hand. (L3, PDF p. 150)

299. a The tape reads calibrated airspeed only, so a marker on it is a CAS value whatever the altitude or active reference. (L3, PDF p. 150)

300. b Synchronizing to the actual speed at activation stops the controller making a violent correction. (L3, PDF p. 150)

301. a For an extended flap, slat and gear configuration the printed rule takes whichever of VFE and VLE is the lower. (L3, PDF p. 150)

302. c The crossover pushbutton beside the speed and Mach display allows a change over at any altitude. (L3, PDF p. 152)

303. b Above the change over the Mach number is held and below it the indicated airspeed is held. (L3, PDF p. 153)

304. b Integrating the lasting error corrects the attitude reference until the error is removed. (L3, PDF p. 154)

305. b The speed error is taken out first, and only then does the aircraft pitch down. (L3, PDF p. 154)

306. b Changing attitude puts the aircraft on the flight path angle that holds the reference speed for the thrust set. (L3, PDF p. 154)